



Reporting Deck OCA Indonesia

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Executive Summary

Project Description :

This project analyzes multi-channel messaging transaction data at OCA Indonesia to build a unified, interactive dashboard that monitors channel performance, delivery effectiveness, user behavior, and operational risks in order to support data-driven decisions for product, business, and management teams.

Objectives :

Create A unified, always-updated dashboard to support operational and business decision-making.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Executive Summary

Tools :

- **EDA** : Big Query, Spreadsheet
- **Visualisation** : Tableau

Metrics : Total Message, Total Charge, Total Failed, Success & Failure Rate, Total Revenue, Channel Performance, User Message Count, Peak Day and Hour.

Analysis Result:

- Whatsapp Channel contributes the highest volume, while SMS is the most stable with lowest failed percentage.
- Call channel is the most lowest volume and highest failed percentage.
- Services dominated by top 3 users.
- 8-5 work hours is actually peak period.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



Business Background

OCA Indonesia is a CPaaS provider under Telkom Indonesia that enables businesses to send large-scale messages across WhatsApp, SMS, Email, and Voice channels, where maintaining delivery reliability, billing accuracy, and cross-channel visibility is critical to sustaining revenue and customer trust.

Note : “This project is part of the RevoU x Telkom Indonesia Virtual Internship Program.”

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Business Problem

Currently, each channel is monitored separately, making it difficult to understand cross-channel performance and user behavior holistically. A unified, always-updated dashboard is needed to support operational and business decision-making.

Objective of analysis

To build an interactive, unified dashboard that provides visibility into multi-channel performance, delivery effectiveness, and user behavior to enable timely, data-driven operational and business decisions.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Methodology

1

Problem Understanding

identify high-risk segments and optimize underwriting decisions.

2

Data Preparation & Data Cleaning

Update created_at and data timestamp to year 20XX from 00XX. Since it's a timestamp, UTC is default.
Conclusion : the time already Machine server timezone.

3

Exploratory Data Analysis

Conduct Initial Investigation on dataset to define metrics and test for hypothesis

4

Data Analysis

Conduct the analysis with the metrics that decided on EDA with SQL and excel.

5

Data Visualization

Visualize the analysis into chart, table, and dashboard by using Tableau

6

Insight & Recommendation

Use the result of data analysis to determined the insight and business recommendation to solve the problem

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Root Cause Analysis

- Uneven delivery reliability across channels.
- Heavy dependency on **WhatsApp** as a primary channel.
- Delivery instability during **peak traffic hours**.
- Volume and revenue concentration among a **small group of high-activity users**.
- Limited early-warning capability without unified monitoring.

Dataset Scope

Table :

- Call : Is_Charge, Created_at, User_id, Total_Price
- Sms : Is_Charge, Created_at, User_id, Total_sent, Total_Price
- Whatsapp : Is_Charge, Created_at, User_id, Price
- Email : Is_Charge, Created_at, User_id, Price
- Users : User_id, User_Name

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



KPI Overview

Total Messages

126,502

Total Charged Count :

100,228 (79.23%)

Failed Count :

26,274(20.77%)

Active Users :

20

Billed:

Rp. 25,946,420

Overall platform activity is **driven by high transaction volume**, with performance stability varying across channels.

KPI overview allows stakeholders to **identify performance degradation early** before deeper investigation.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



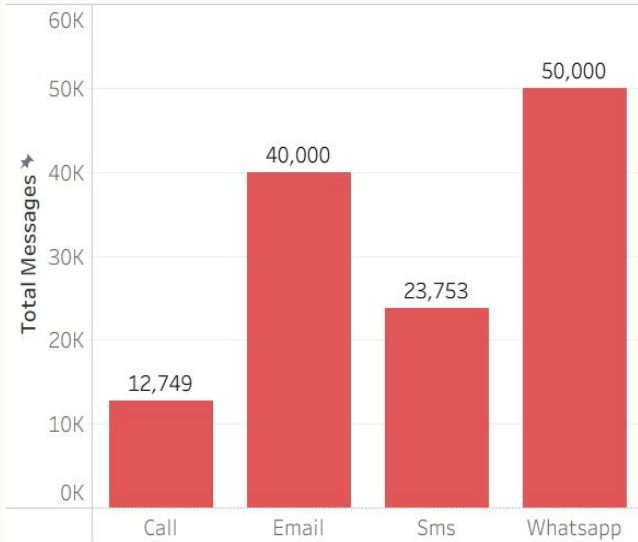
omni
communication
assistant



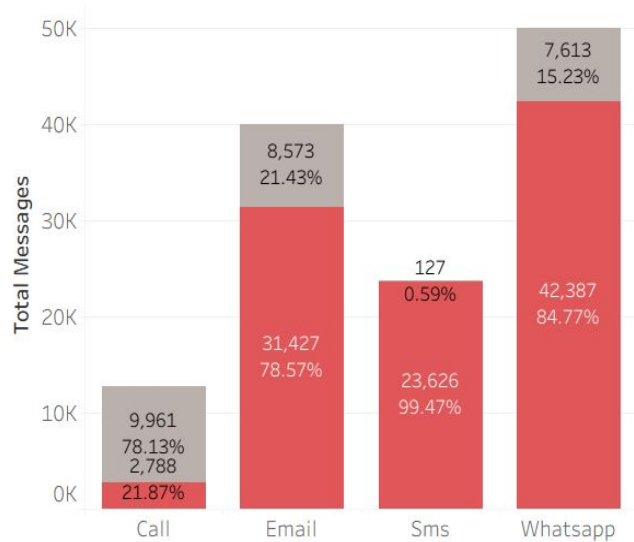
Channel Performance

Channel Performance

Message By Channel



Delivery Trend



WhatsApp contributes the highest message volume and revenue, making it the most critical channel.

SMS shows the **lowest failure rate**, indicating strong delivery stability despite lower volume.

Call channel has the **lowest usage and highest failure rate**, signaling operational inefficiency.

Is Charge

■ Failed
■ Charged

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



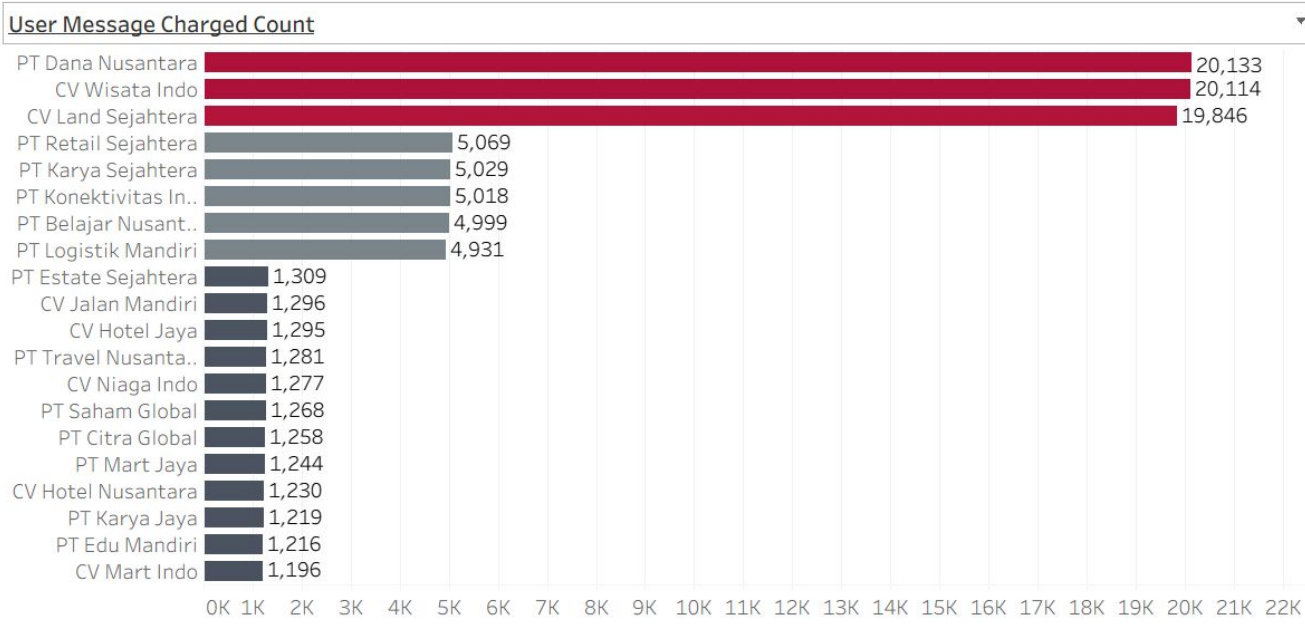
omni
communication
assistant



KPI User Message - Charged

Top 3 User Dominate more than half of the services :

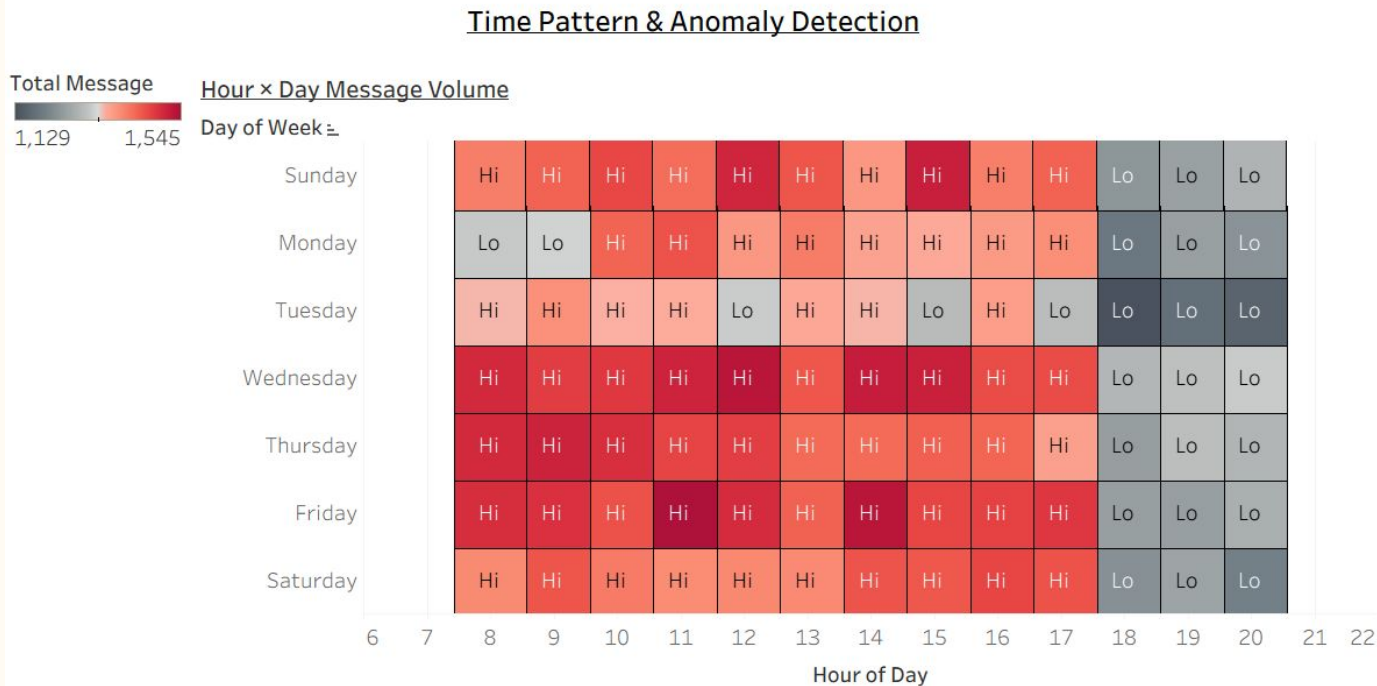
PT. Dana Nusantara
CV Wisata Indo
CV Land Sejahtera



Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



Time Pattern and anomaly Detection



Message volume peaks during **specific business hours and weekdays**.

Delivery effectiveness does **not always increase proportionally** with message volume.

Failure rates show **spikes at certain hours**, indicating potential congestion or gateway constraints.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Insight And Recommendations

- Diversify client usage across channels to reduce **mono-channel dependency risk**.
- Strengthen **peak-hour monitoring and alerting** to prevent SLA breaches.
- Prioritize stability improvements for **high-volume and high-revenue channels**.
- Use time-based anomaly detection as an **early-warning system** for Ops.
- Leverage user behavior insights to identify **upsell and retention opportunities**.
- Dashboard Filters :
 - User : Allows focused analysis of individual user activity to identify high-impact users
 - Channel: Enables comparison of performance and reliability across communication channels.
 - Lookback Days: Provides dynamic monitoring of recent performance trends based on the latest available data.

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



the world in your hand



[You can check the tableau public here](#)

[Connect to My LinkedIn](#)



Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



Appendix - SQL Query

```
select tr.Type, tr.transaction_id, tr.user_id, us.user_name,us.field_of_business, us.join_date,
tr.is_charge, tr.message_type,tr.total_sent, tr.total_price, date_add(datetime(tr.created_at), interval 2000 year) created_at,
tr.message_status,
date_add(datetime(tr.received_timestamp), interval 2000 year) received_timestamp -- the timestamp have lacking 2000 years due
to convert data. while timestamp always saved as UTC, the real timezone is from the real server of channel.
from
(SELECT "Call" Type, transaction_id, user_id, is_charge, "Call" message_type, 1 total_sent, total_price, created_at,
message_status, created_at as received_timestamp FROM `revou-vi-001.telkom_oca.Call`
union all
SELECT "Email" Type,transaction_id,user_id, is_charge, "Email" message_type, 1 total_sent, price, created_at, message_status,
created_at as received_timestamp FROM `revou-vi-001.telkom_oca.Email`
union all
SELECT "Sms" Type,transaction_id,user_id, is_charge, "Sms" message_type, total_sms as total_sent, total_price, created_at,
message_status, created_at as received_timestamp FROM `revou-vi-001.telkom_oca.Sms`
union all
SELECT "Whatsapp" Type,transaction_id,user_id, is_charge, message_type, 1 total_sent, price, created_at, last_status as
message_status, received_timestamp FROM `revou-vi-001.telkom_oca.Whatsapp`) tr
left join `revou-vi-001.telkom_oca.Users` us on tr.user_id = us.user_id
```

Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."



omni
communication
assistant



Appendix - Virtual Internship Pitching Video



Disclaimer : "The data provided is a representation of the original data which amount has been adjusted. The data used is NOT representation of original data, adjusted for educational purposes and does not reflect the actual business condition of the company."

